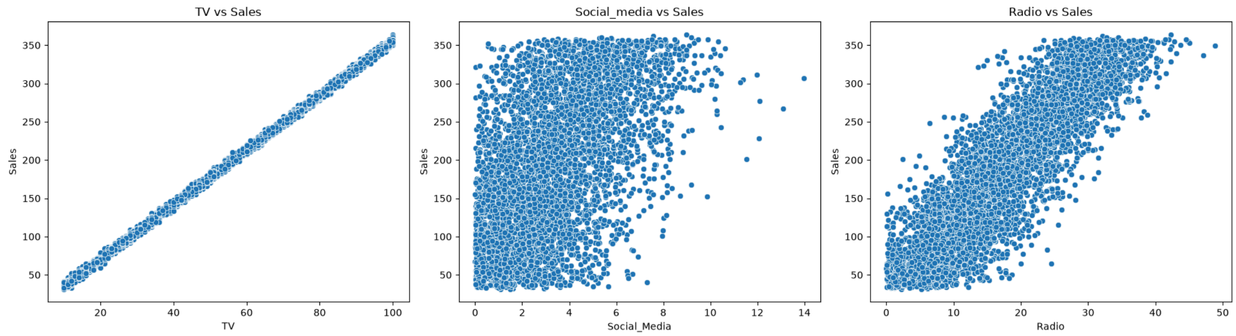
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
import scipy.stats as stats

df = pd.read_csv('regression.csv')
df.head()
df.info()
df.describe()
```

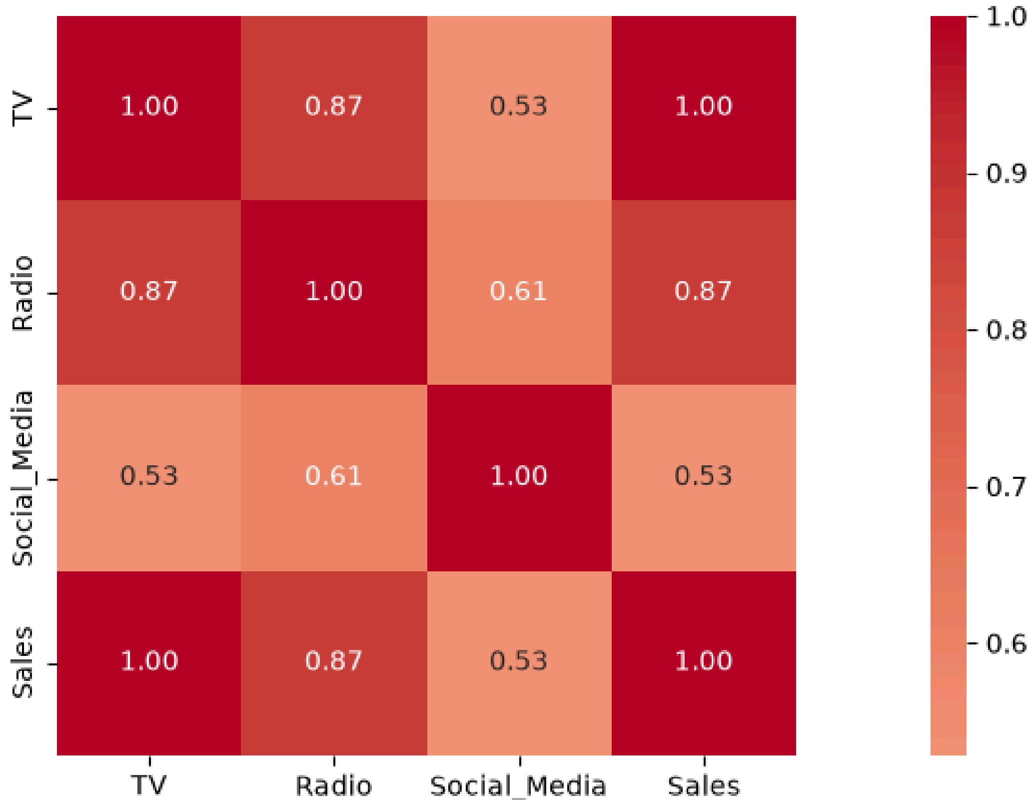
```
<class 'pandas.DataFrame'>
RangeIndex: 4572 entries, 0 to 4571
Data columns (total 4 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   TV            4562 non-null   float64
 1   Radio         4568 non-null   float64
 2   Social_Media  4566 non-null   float64
 3   Sales         4566 non-null   float64
dtypes: float64(4)
memory usage: 143.0 KB
```

	TV	Radio	Social_Media	Sales
count	4562.000000	4568.000000	4566.000000	4566.000000
mean	54.066857	18.160356	3.323956	192.466602
std	26.125054	9.676958	2.212670	93.133092
min	10.000000	0.000684	0.000031	31.199409
25%	32.000000	10.525957	1.527849	112.322882
50%	53.000000	17.859513	3.055565	189.231172
75%	77.000000	25.649730	4.807558	272.507922
max	100.000000	48.871161	13.981662	364.079751

```
plt.rcParams['figure.figsize'] = (18,5)
fig, axes = plt.subplots(1,3, figsize=(18,5))
sns.scatterplot(data=df, x='TV', y='Sales', ax=axes[0])
sns.scatterplot(data=df, x='Social_Media', y='Sales', ax=axes[1])
sns.scatterplot(data=df, x='Radio', y='Sales', ax=axes[2])
axes[0].set_title('TV vs Sales')
axes[1].set_title('Social_media vs Sales')
axes[2].set_title('Radio vs Sales')
plt.tight_layout()
plt.show()
```




```
corr = df[['TV','Radio','Social_Media','Sales']].corr()
sns.heatmap(corr,annot=True, cmap='coolwarm', center=0, fmt='.2f', square=True)
plt.show()
print(corr['Sales'].sort_values(ascending=True))
```



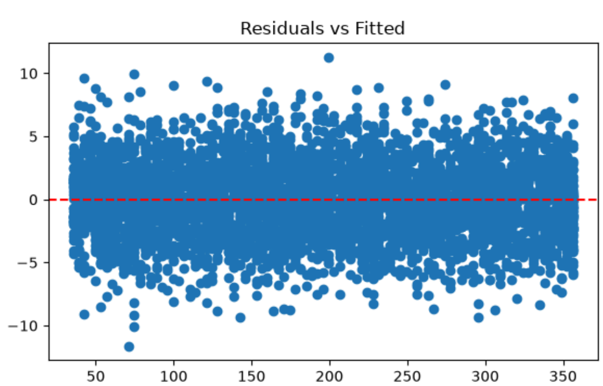

```
Social_Media    0.528906
Radio           0.869105
TV              0.999497
Sales           1.000000
Name: Sales, dtype: float64
```

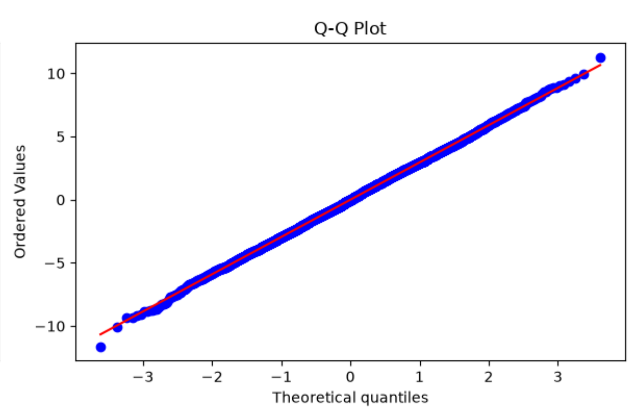
```
df_clean = df[['TV','Sales']].dropna()
X = df_clean['TV']
y = df_clean['Sales']
X = sm.add_constant(X)
model = sm.OLS(y,X).fit()
print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  Sales   R-squared:                       0.999
Model:                            OLS   Adj. R-squared:                  0.999
Method:                 Least Squares   F-statistic:                 4.527e+06
Date:                Sat, 13 Jun 2026   Prob (F-statistic):               0.00
Time:                        04:15:39   Log-Likelihood:                -11393.
No. Observations:                4556   AIC:                         2.279e+04
Df Residuals:                    4554   BIC:                         2.280e+04
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const         -0.1263      0.101     -1.257      0.209      -0.323       0.071
TV             3.5614      0.002   2127.776      0.000       3.558       3.565
==============================================================================
Omnibus:                        0.051   Durbin-Watson:                   2.002
Prob(Omnibus):                  0.975   Jarque-Bera (JB):                0.030
Skew:                           0.001   Prob(JB):                        0.985
Kurtosis:                       3.012   Cond. No.                         138.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spe
```

```python
residuals = model.resid
fitted = model.fittedvalues
fig, axes = plt.subplots(1,2, figsize=(12,4))
axes[0].scatter(fitted, residuals)
axes[0].axhline(0, color='r', linestyle='--')
axes[0].set_title('Residuals vs Fitted')
stats.probplot(residuals, dist="norm", plot=axes[1])
axes[1].set_title('Q-Q Plot')
plt.tight_layout()
plt.show()
```





Interpretation

R-squared = 0.999: 99.9% of sales variation explained by TV spend. TV coefficient = 3.5614, p < 0.001: Statistically significant positive impact. F-statistic confirms model is highly significant.

Business Recommendation

Allocate majority of marketing budget to TV channel. Each additional unit spent on TV drives predictable increase in Sales. Test TV + Radio combination next to capture remaining 0.1% variation.